

PerinataIKG — ETL Pipeline and Knowledge Graph Construction

DATA SOURCES

SINASC

Live births
DBC files

SIM

Neonatal deaths
(linkage validation)

BR-DWGD

Gridded climate
0.1° × 0.1°

IBGE

Municipalities
centroids + HDI

STAGE 1
DBC → Parquet

R / microdatasus
DBC → CSV
Polars (Parquet)

Harmonisation
Date normalisation

STAGE 2
Climate Linkage

Spatial join:
centroid → BR-DWGD cell
(nearest neighbour)
Trimester assignment:
T1: wk 1–12 T2: wk 13–26
T3: wk 27 – birth

STAGE 3
RDF Conversion

rdflib 6.x (Python)
~14 triples / birth
OWL class typing
PostTermBirth: IG ≥ 42w
ExtremeHeat flag:
Tmax > 35°C

STAGE 4
Fuseki Load

Apache Jena Fuseki 6.0.0
TDB2 persistent store
HTTP POST (batch)
TDB2 deduplication
(1 duplicate resolved)

SPARQL
ENDPOINT

Triples
13,334,157
Throughput
4,439 / s
Query range
0.1 – 7.6 s

Births
934,806
Load time
~119 s
Software
Fuseki 6.0.0

Pipeline notes:

- 934,806 distinct Birth individuals after TDB2 deduplication (1 duplicate birth_id across annual Parquet files)
- Pipeline throughput: 4,439 births/s | Conversion: ~211 s | TDB2 load: ~119 s | Total: ~330 s

- Extreme heat (ExtremeHeatExposure): Tmax > 35°C on ≥ 1 day during T3 (absolute threshold, uniform across municipalities)
- Hardware: AMD Ryzen 9 3900X (24 threads), 64 GB RAM, Ubuntu Linux | Dataset: Goiás state, Brazil, 2015–2024